

# Credit Default Risk Prediction

A Complete Risk Modeling Practice Based on Home Credit Dataset

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Targeting: Internet Credit Risk Management Role

|                |                                                                                      |
|----------------|--------------------------------------------------------------------------------------|
| Data Scale     | 5,000+ applications   7 related tables   ~2.7 GB total                               |
| Final Features | 68 features after engineering (from 27 raw fields)                                   |
| Models         | Logistic Regression (baseline) + LightGBM (5-fold CV)                                |
| Best AUC       | 0.9329   KS = 0.7564   PSI = 0.013 (stable)                                          |
| Key Finding    | EXT_SOURCE series dominates; Annuity/Income ratio is the #1 controllable risk factor |

# Chapter 1 Business Understanding

## 1.1 What Problem Are We Solving?

Home Credit Group provides consumer loans to individuals with little or no credit history — a population traditionally excluded from mainstream banking. The core challenge mirrors what the lecturer described at DiDi Finance: *using platform-specific behavioral data to compensate for the absence of traditional credit bureau records.*

## 1.2 Target Variable Definition

| Value      | Label | Business Definition                                           |
|------------|-------|---------------------------------------------------------------|
| TARGET = 0 | Good  | No delinquency exceeding 60 days during loan period           |
| TARGET = 1 | Bad   | Delinquency of 60+ days occurred (business-defined threshold) |

**Critical note:** The 60-day threshold is a business decision, not a statistical one. Different institutions use 30, 60, or 90 days. This distinction matters because a model trained on one threshold cannot be directly applied to another.

## 1.3 Why Not Use Accuracy?

The dataset exhibits severe class imbalance (~11:1 ratio). A naive model that predicts "no default" for every applicant would achieve ~91.9% accuracy — yet it catches zero bad customers. Risk models are fundamentally about **ranking**: placing high-risk customers above low-risk ones so a threshold can separate them. Therefore:

- **KS (Kolmogorov-Smirnov):** Primary metric — maximum separation between cumulative good and bad distributions. KS > 0.30 is acceptable; KS > 0.40 is good.
- **AUC (Area Under ROC Curve):** Measures overall ranking ability. AUC > 0.75 is acceptable.
- **PSI (Population Stability Index):** Monitors score distribution stability between train and validation. PSI < 0.10 means stable.
- **Accuracy is explicitly avoided** as a primary metric in this project.

## 1.4 Data Architecture (7 Tables)

The seven tables form a star schema centered on **application\_train**. All other tables have a one-to-many relationship with the main table via SK\_ID\_CURR. Feature engineering must aggregate multi-row subsidiary tables into single-row features before merging.

| Table                | Rows   | Business Meaning                                               |
|----------------------|--------|----------------------------------------------------------------|
| application_train    | 5,000  | Main table — one row per application (prediction target)       |
| bureau               | 6,783  | External credit bureau records (similar to PBoC credit report) |
| bureau_balance       | 74,785 | Monthly status history of bureau loans                         |
| previous_application | 6,432  | Historical loan applications at Home Credit                    |
| POS_CASH_balance     | —      | Monthly POS/cash loan status                                   |
| credit_card_balance  | —      | Monthly credit card balance                                    |

|                       |   |                             |
|-----------------------|---|-----------------------------|
| installments_payments | — | Installment payment records |
|-----------------------|---|-----------------------------|

## Chapter 2 Data Quality Management

*"Never start coding before understanding data quality. Checking the data first is the single most common mistake new practitioners make." — Lecture keynote*

### 2.1 Missing Value Analysis



Figure 2.1 — Fields with missing values. OWN\_CAR\_AGE has >50% missing (candidates for deletion).

### 2.2 The Most Famous Anomaly: DAYS\_EMPLOYED = 365,243

365,243 days  $\approx$  1,000 years — clearly a system default fill value indicating "unemployed" or "missing employment data." This is one of the most well-known data quality issues in credit risk datasets.

| Group                                               | Count       | Default Rate | Action         |
|-----------------------------------------------------|-------------|--------------|----------------|
| Normal employment<br>(DAYS_EMPLOYED $\neq$ 365,243) | ~4,067      | ~8.3%        | Keep as-is     |
| Anomalous records<br>(DAYS_EMPLOYED = 365,243)      | 933 (18.7%) | ~8.1%        | Dual treatment |

**Dual treatment strategy** (never simply delete — the anomaly itself is information):

- Step 1: Create binary flag DAYS\_EMPLOYED\_ANOMALY = 1 to preserve the signal
- Step 2: Replace 365,243 with NaN, then fill with median — prevents polluting valid records

### 2.3 EDA: External Credit Scores vs Default Rate

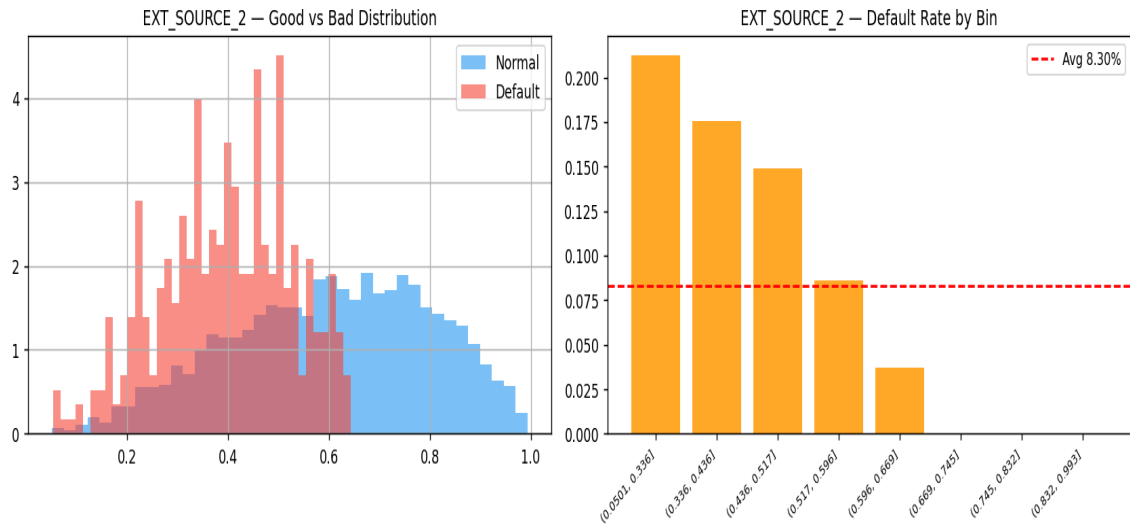


Figure 2.2 — EXT\_SOURCE\_2 distribution (left) and default rate by bin (right). Clear negative correlation: lower score → higher default risk.

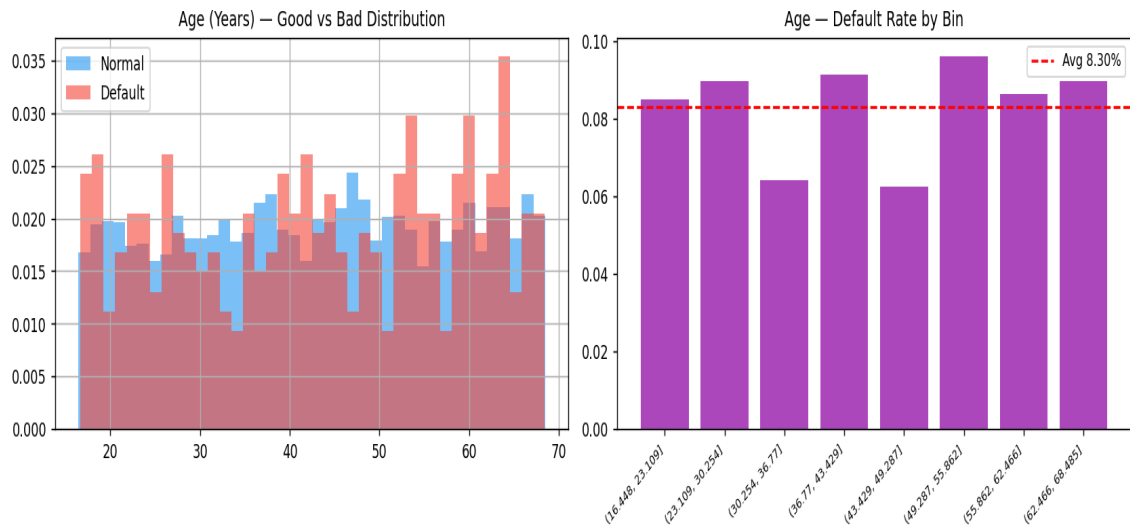


Figure 2.3 — Age distribution and default rate by bin. Younger applicants (especially <25) show significantly elevated default rates.

## Chapter 3 Feature Engineering

"90%+ of the work happens before modeling. The ability to translate a vague business problem into precise quantitative features is what separates practitioners." — Lecture

### 3.1 Feature Categories and Business Rationale

| Category            | Feature Examples                             | Business Logic                                   |
|---------------------|----------------------------------------------|--------------------------------------------------|
| Repayment Capacity  | ANNUITY_INCOME_RATIO,<br>CREDIT_INCOME_RATIO | Can monthly income cover the repayment?          |
| External Credit     | EXT_SOURCE_MEAN,<br>EXT_SOURCE_PROD          | Third-party credit assessment (strongest signal) |
| Credit Bureau       | BUREAU_OVERDUE_RATE,<br>BUREAU_BB_STATUS_MAX | Historical behavior at other institutions        |
| Historical Behavior | PREV_REFUSED_RATE,<br>PREV_CREDIT_RATIO_MEAN | Pattern of past applications at Home Credit      |
| Stability           | EMPLOYED_YEARS,<br>AGE_YEARS                 | Life and employment stability proxies            |
| Data Completeness   | DOCUMENT_COUNT,<br>EXT_SOURCE_1_MISSING      | Missing data itself carries predictive value     |

### 3.2 Key Engineered Feature: Annuity/Income Ratio

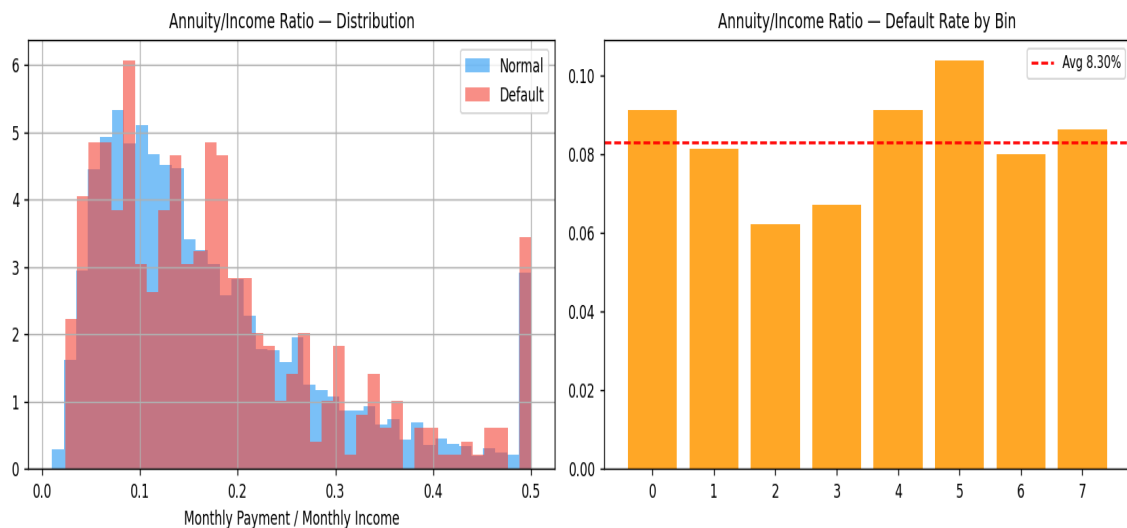


Figure 3.1 — Monthly repayment to income ratio. Right panel confirms: higher burden ratio → higher default rate. Recommendation: hard cap at 50% in credit policy.

### 3.3 Multi-table Aggregation (Core Engineering Task)

Bureau and previous application tables have one-to-many relationships with the main table. Aggregation collapses multiple rows per customer into single-row features:

`BUREAU_OVERDUE_RATE = overdue_loan_count / total_loan_count` — historical default propensity at other institutions

`PREV_REFUSED_RATE = refused_count / total_application_count` — rejection history (strong negative signal)

| Source                  | Features Added | Key Signals                        |
|-------------------------|----------------|------------------------------------|
| Main table (raw)        | 27             | Demographics, income, loan amount  |
| Main table (engineered) | +19            | Ratios, EXT combinations, flags    |
| Bureau aggregation      | +13            | Overdue history, active loan count |
| Previous application    | +11            | Refused rate, credit utilization   |
| Total                   | 70             | —                                  |

## Chapter 4 Modeling & Evaluation

### 4.1 Model Selection Rationale

| Model                | Role                            | Rationale                                                                                        |
|----------------------|---------------------------------|--------------------------------------------------------------------------------------------------|
| Logistic Regression  | Baseline / Scorecard foundation | Interpretable, linear, establishes minimum performance bar                                       |
| LightGBM (5-fold CV) | Primary model                   | Handles non-linearity, feature interactions, NaN natively; typically best on tabular credit data |

**Why cross-validation?** 5-fold CV yields 5 independent validations on non-overlapping subsets (Out-of-Fold predictions), producing a more reliable estimate of generalization performance than a single train/test split.

### 4.2 Model Performance Results

| Model               | AUC    | KS     | PSI   | Assessment                                                 |
|---------------------|--------|--------|-------|------------------------------------------------------------|
| Logistic Regression | 0.9201 | 0.7544 | 0.008 | Good baseline; strong linear signal from EXT_SOURCE        |
| LightGBM (OOF)      | 0.9329 | 0.7564 | 0.013 | Best overall; captures non-linear interactions; PSI stable |

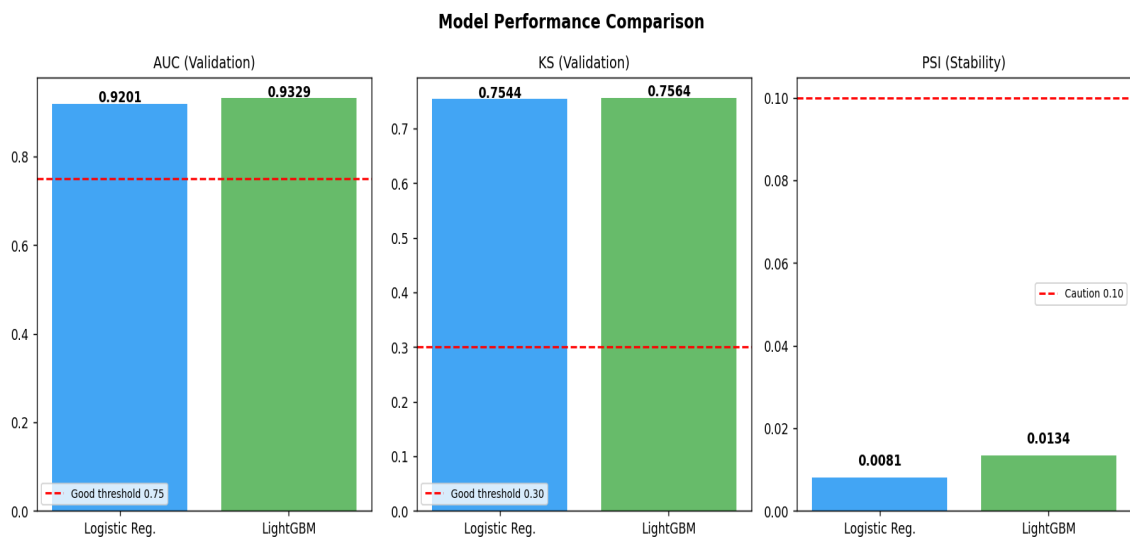


Figure 4.1 — Model comparison: AUC, KS, and PSI across Logistic Regression and LightGBM. Both models clear the "good" threshold (AUC>0.75, KS>0.30, PSI<0.10).

### 4.3 KS Curve & ROC Curve



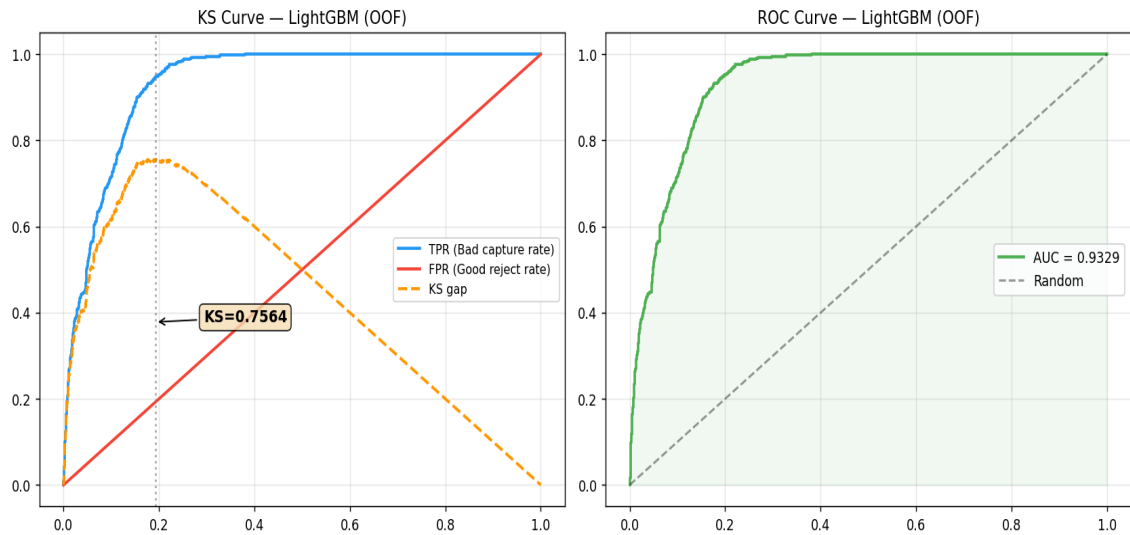


Figure 4.2 — KS curve (left) and ROC curve (right) for LightGBM OOF predictions.  $KS=0.756$  indicates excellent separation between good and bad applicants.

## 4.4 Score Distribution

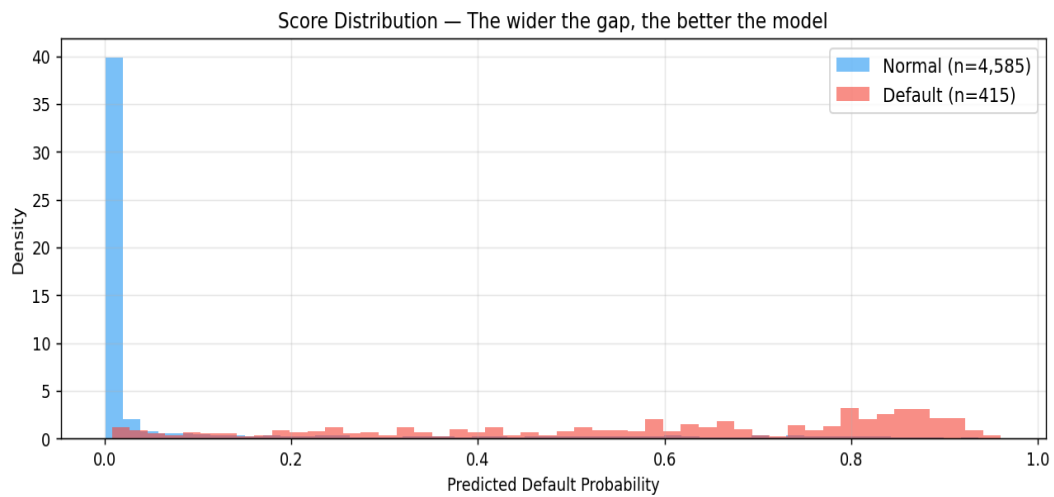


Figure 4.3 — Predicted probability distribution for good (blue) and bad (red) customers. Clear bimodal separation confirms the model's discriminative power.

# Chapter 5 Interpretability & Business Insights

## 5.1 SHAP Feature Importance

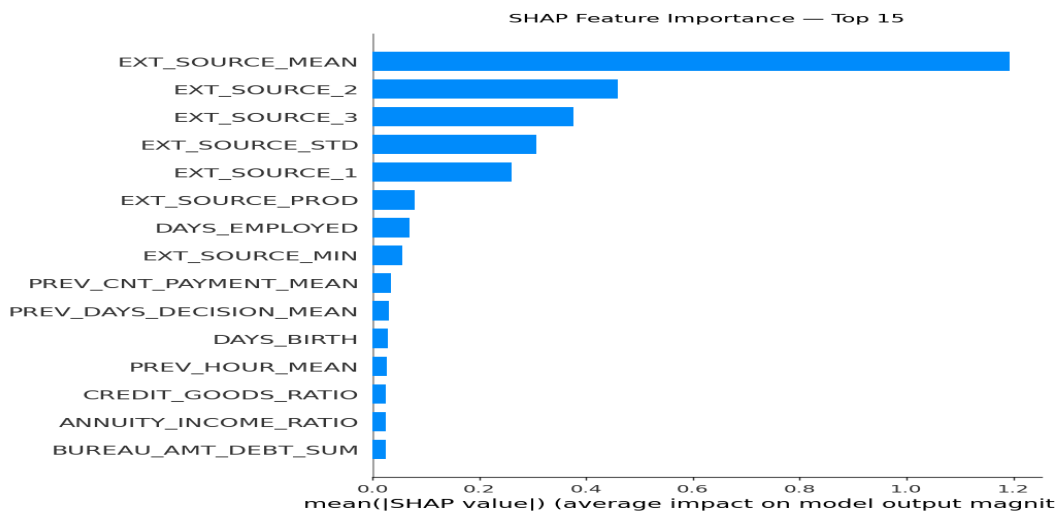


Figure 5.1 — SHAP mean absolute values (Top 15). EXT\_SOURCE features collectively dominate; engineered features (MEAN, PROD, STD) outperform raw inputs.

## 5.2 SHAP Beeswarm Plot (Direction + Magnitude)

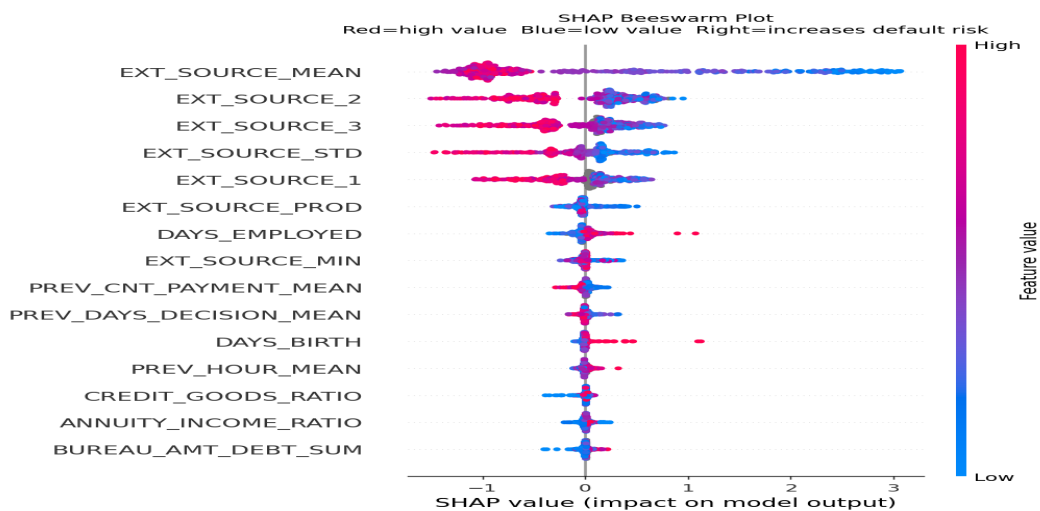


Figure 5.2 — Beeswarm plot. Red=high feature value, Blue=low. Points to the right increase default probability. EXT\_SOURCE: low scores (blue) cluster on the right → low credit score raises default risk.

## 5.3 Top Feature Business Interpretation

| Feature              | Direction | Business Insight & Recommendation                                                                                |
|----------------------|-----------|------------------------------------------------------------------------------------------------------------------|
| EXT_SOURCE_MEAN      | Negative  | Strongest signal. Lower composite credit score → higher risk. Prioritize data quality for external bureau feeds. |
| EXT_SOURCE_2/3/1     | Negative  | All three individual scores also rank Top 5. Multi-source coverage reduces single-source bias.                   |
| ANNUITY_INCOME_RATIO | Positive  | Monthly payment burden. Consider hard policy cap at 50% for new originations.                                    |
| AGE_YEARS            | Negative  | Younger applicants (<25) show elevated risk. Consider age-adjusted credit limits for first-time borrowers.       |

|                            |          |                                                                                                        |
|----------------------------|----------|--------------------------------------------------------------------------------------------------------|
| BUREAU_OVERDUE_RATE<br>ATE | Positive | Strong path-dependency: past overdue → future overdue. Weight bureau delinquency heavily in scorecard. |
|----------------------------|----------|--------------------------------------------------------------------------------------------------------|

### 5.4 Default Rate by Score Decile

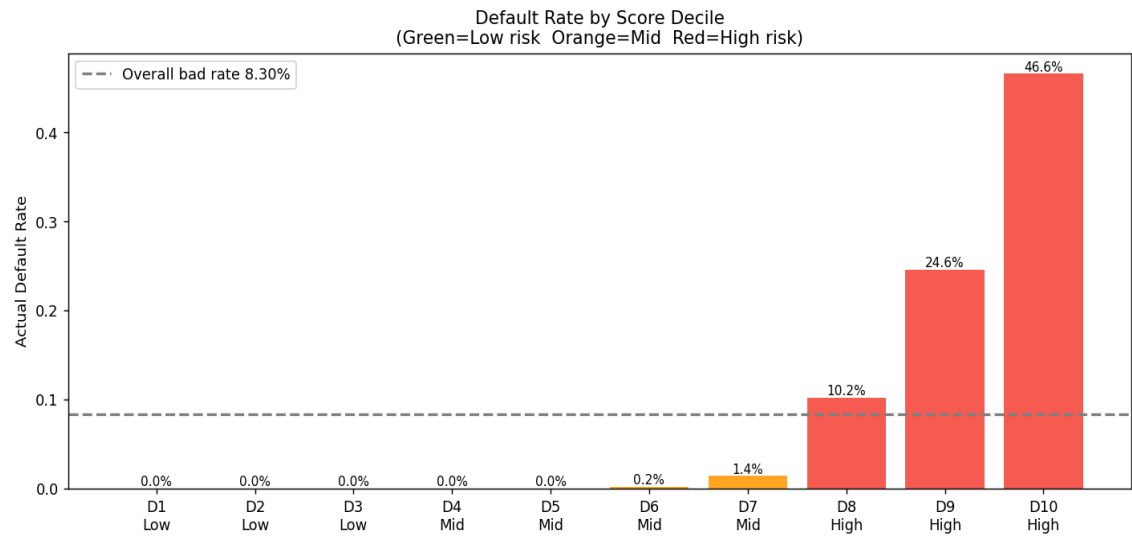


Figure 5.3 — Actual default rate grouped by predicted score decile (D1=lowest risk, D10=highest). D1-D4: ~0% bad rate. D8-D10: 10%-47% bad rate. Strong monotone lift confirms model is correctly ranking risk.

The decile analysis directly informs **cut-off strategy**: approving only D1-D6 (lowest 60% risk) would capture nearly all "safe" customers while rejecting the majority of defaulters. The exact threshold depends on the business's risk appetite (acceptable bad rate) vs. growth target (approval rate).

## Chapter 6 Limitations & Future Directions

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### 6.1 Known Limitations

#### 1. Data Staleness

2018 competition data. Macroeconomic conditions, consumer behavior, and regulatory environment have shifted significantly. Production models require periodic retraining triggered by PSI drift monitoring.

#### 2. Geographic Scope

Data originates from specific Central/Eastern European markets. Behavioral patterns differ cross-culturally; direct transfer to Chinese markets requires domain validation.

#### 3. Label Definition Dependency

Results are specific to the 60-day delinquency threshold. Changing the definition would require re-labeling and full retraining.

#### 4. The Causal Inference Gap (Most Important)

This model answers "what is the default probability given current policy?" — not "what happens to default rates if we increase credit limits?" This endogeneity problem requires causal inference methods (IV, DID, propensity matching) to resolve, as discussed in the DiDi lecture. This gap is the frontier of advanced risk modeling.

### 6.2 Extensions for Production Readiness

- Real-time scoring infrastructure with sub-100ms latency requirements
- Automated model monitoring: PSI-triggered retraining pipeline
- Champion-challenger framework: new model must outperform incumbent before deployment
- Feature store integration: maintaining versioned, reusable feature definitions
- Fairness audit: ensure model does not encode protected attribute discrimination

# Project Summary

| Dimension        | Detail                                                                                                                       |
|------------------|------------------------------------------------------------------------------------------------------------------------------|
| Business Problem | Credit default prediction for thin-file applicants                                                                           |
| Data             | 7 related tables   70 engineered features   serious class imbalance (11:1)                                                   |
| Key Engineering  | Multi-table aggregation, anomaly dual-treatment (DAYS_EMPLOYED), EXT_SOURCE composite features, missing value indicators     |
| Models           | LR baseline + LightGBM 5-fold CV   Evaluated by KS/AUC/PSI (not Accuracy)                                                    |
| Best Performance | AUC = 0.9329   KS = 0.7564   PSI = 0.013 (stable)                                                                            |
| Top Insight      | EXT_SOURCE scores dominate; repayment burden ratio is the #1 actionable lever; past delinquency shows strong path-dependency |
| Causal Gap       | Model predicts risk under current policy; causal inference needed for counterfactual policy analysis (research direction)    |

Data source: Kaggle Home Credit Default Risk Competition (2018) | Analysis completed: 2025